

Let $X \subseteq \mathbb{R}^n$ be a subset of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$ an interior point of X . It means that there exists $\varepsilon > 0$ such that

$$\|z - x\| \leq \varepsilon \implies z \in X.$$

Show that any direction $d \in \mathbb{R}^n$ is feasible in x .